Skills Summary

Zeros That Hold a Place

Use this reference while completing your worksheet or when studying.

Skill 1 — What a zero does in a number

The model: every number has three places.

hundreds	**tens**	**ones**
100s	10s	1s

A 0 means *none of that size* — and it holds the place open so the other digits do not slide over.

Example: 5 hundreds, 0 tens, 3 ones.
Step 1. Give each place its own piece, including the empty one, so nothing slides.
Step 2. $500 + 0 + 3$, and the pieces make **503**.

Try it: 8 hundreds, 0 tens, 6 ones. What number is it?

check: **806**

Skill 2 — Fixing a missing zero

Check your work like this: add up the pieces of your expanded form. If the total is not the number you started with, a digit has slid into the wrong place. Read the number again, one place at a time, and give the empty place its 0.

Example: Sam writes $604 = 600 + 40$. Fix it.
Step 1. Add Sam's pieces: $600 + 40 = 640$, which is not 604, so a digit slid — check each place.
Step 2. 604 has 6 hundreds, 0 tens, 4 ones, so it is $600 + 0 + 4 = \mathbf{604}$. The 4 is ones, not tens.

Try it: Fix this: $209 = 200 + 90$. What should the pieces add up to?

check: **209**

Skill 3 — Same digits, different value

Rule: to compare, look at the hundreds place first. If the hundreds match, compare the tens. If the tens match too, compare the ones. A 0 in a place means that place is worth nothing — so it is the smallest a place can be.

Example: Compare 507 and 570.
Step 1. Both have 5 hundreds, so the hundreds do not decide it — move to the tens place.
Step 2. 507 has 0 tens and 570 has 7 tens, so $\mathbf{507 < 570}$.

Try it: Write $<$ or $>$ between 830 and 803.

check: **830 $>$ 803**

(!) Watch out: Do not skip the empty place when you write expanded form. 308 is $300 + 0 + 8$. If you write $300 + 80$ you have moved the 8 into the tens place and made a different number, 380.